Employing Image Processing Techniques for Hardware Trojans Detection

Mohammad Mehdi Rahimifar
Dept. of Electrical Engineering
University of Kurdistan
Sanandaj, Kurdistan, Iran
m.m.rahimifar@eng.uok.ac.ir

Hadi Jahanirad
Dept. of Electrical Engineering
University of Kurdistan
Sanandaj, Kurdistan, Iran
h.jahanirad@uok.ac.ir

***Abstract*— Hardware Trojans (HT) become an emerging issue in the integrated circuits (IC) production, which have many malicious effects on the ICs such as functionality changes, overheating problems and leaking information. According to the Trojans' small size and ICs complex structure, detecting Trojans is a serious challenge to solve. In this paper, we propose a novel approach in order to detect very small Trojans with high accuracy. First, we generate the ideal power heatmap of ISCAS benchmarks with calculating switching activities and the dynamic power of each gate and node of the testing circuits. Second, we employ image processing techniques (similarity structure index and thresholding) to detect differences in the heatmaps which caused by Trojans. With our approach, very small Trojans that consume power less than 0.001 of the total power consumption can be detected and located with 97 percent average accuracy.**

Keywords— Hardware Trojans, Image processing, Power heatmap, ASIC, Switching activity, Power estimation, Structural similarity index, Thresholding

I. INTRODUCTION

Hardware Trojans can downgrade the IC performance, change its functionality and cause denial of service and the current leakage problem [1], which may increase IC's dynamic power and overheating in the different parts of integrated circuits [2]. Unfortunately, Trojans Detection is a very difficult challenge due to the differences between Trojans' sizes. The most efficient way to find Hardware Trojans is the reverse engineering [3]. Despite the reverse engineering benefit, which is great reliability on Trojans detection, it may cause massive destruction and expense. Since Hardware Trojans have unwanted outcomes on the power and temperature of circuits, many methods have been developed to detect for trojans such as thermal scanning and machine learning [4-9]. In this paper, we focus on a stable stage in the long-term run-time of the IC. Hardware Trojans can always be on or be triggered to activate, and they can be inserted in many different levels 1- design, 2- fabrication, 3- testing, 4- assembly [1,2]. There are previous works on Hardware Trojans detection using machine learning and thermal scanning methods through heatmaps. In [6], the authors developed a novel technique to detect Trojans in thermal maps and inversed power maps using 2DPCA (two-dimensional principal component analysis with machine learning techniques (supervised and unsupervised). In [7], the authors proposed approaches for run-time Trojan detection to detect a deviation in power and thermal profiles with a local sensor-based approach. In [9], the authors proposed the SSIM (Structural similarity index) algorithm to compare a SEM (scanning electron microscope) image from a backside golden IC and low-quality SEM image of an IC under authentication to detect changed parts of the circuit and find Trojans' locations. The major drawbacks with the previous works are 1- some factories may change and optimize the circuit layout to increase the yield, so scanning the layouts using the SSIM, may have some errors, 2- in the SEM imaging and thermal map scanning, image and thermal noises cause errors in the final result. Comparing with the previous works, we generate an accurate power heatmap with no noise for the testing circuits using the switching activity method, to create an ideal heatmap. We propose the thresholding and the SSIM algorithms [10] to detect Trojans in the Trojan included heatmaps. In our approach, we can detect Trojans that are inserted after the design phase and have negligible effects on the power. The heatmaps provided from the dynamic power of circuit gates can cover the lack of thermal and power heatmaps of the ICs from the factories. Power consumption using the switching activity technique, can compute the dynamic power of each logic gate in the circuit [11-14]. We have designed an architecture to place our testing circuits logic gates in the ASIC. This means that each ASIC grids (blocks) only contain one logical gate. With this approach of placement, switching activity, capacitance, and the dynamic power of each gate can be computed respectively. Accordingly, with the calculation of the dynamic power of logic gates (which are placed in each ASIC blocks), we can generate heatmaps based on the dynamic power matrix. These heatmaps can provide precise power estimation of each gate of the circuit and supply the ideal power heatmaps without any Trojans included. We examine heatmaps to detect Trojans' locations using image processing techniques such as SSIM and the thresholding. Consequently, Hardware Trojans that affect the functionality of the circuit and cause overheating in some different parts of the circuit can be determined. The main contributions of the paper are:

- Generating the placed circuit's ideal power heatmap using the switching activity method.
- Implementing various types of Hardware Trojans in different locations of the testing circuits and providing secondary heatmaps for each Trojan included circuits.
- Employing image processing algorithms for the generated power heatmaps to perform Hardware Trojans detection.

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The remainder of this paper is organized as follows. Section II describes the estimation of the circuits' dynamic power and provides an ideal power map. We also describe image processing algorithms in this section. In Section III, overall reliability evaluation and simulation results are presented. Finally, this paper is concluded in Section IV.

II. METHODS

A. Problem definition

Generating dynamic power heatmaps needs an accurate power model simulation, which is a challenging problem. Many of the previous works proposed methods to detect Trojans without any ideal heatmap, but as the functionality of chips may be different in any conditions, Trojan detection accuracy can be better with evaluating testing circuits with ideal heatmaps. The power heatmaps can be created by a dynamic power matrix. In order to the justification of ideal heatmaps, there must be only one dynamic power value for individual items of matrix (each block of the chips). If a matrix has M rows and N columns, then dynamic powers for filling the matrix must not be over $M \times N$ values. For example, if a chip contains 100 logical gates, it is possible to provide a $M \times N$ matrix ($M \times N = 100$) to illustrate logical gates powers. Dynamic power matrix is defined as:

$$\text{Dynamic Power} = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix} \quad (1)$$

Where each a_{mn} value must be replaced with specific dynamic power according to the components and the logic gates location in the IC, to present an ideal heatmap. We use ISCAS 85 and ISCAS 89 circuits benchmarks to apply our approach for Providing the power heatmaps. Each circuit from ISCAS benchmarks has a specific logic gates number. Table I shows the ISCAS benchmarks characteristics. There are also BLIF (Berkeley logic interchange format) and Verilog HDL (Hardware description language) files available in the benchmark's documents. The ISCAS c17 circuit schematic Is illustrated in Fig. 1. We can observe that there are six logic gates in the c17 circuit. It is a challenging problem to place the ISCAS testing circuits logic gates manually in specific locations to generate ideal matrices. In order to generate heatmaps we have used Reg Place: A High-Quality Structured ASIC Placement Tool to ASIC design [15] for our approach. We perform 2-D placement in the ASIC in order to have a better view of logic gates placement. Fig. 2 provides information about our approach of placement architecture in order to place the ISCAS c17 circuit in a 4×4 ASIC area. We have divided ASIC to separated grids in order to place the testing circuit logic gates in any grid. A 2-D placement of the gates can be provided with this method of placement. In our approach, each grid must not have over one logic gates but can be empty. Although our approach of placement is not the best tactic to place logic blocks (place only one gate in any grid), but we can generate a power matrix and, consequently, the power heatmaps to cover the lack of heatmaps from factories. In the ideal placement of the ASICS, many blocks may have more than one logic gate implemented or be empty. There are empty spaces available in the grids and the routing to place Trojans by designers. According to the logic - gates number in the ISCAS circuits, ASIC area and consequently, the matrices sizes will increase. Estimating the average power dissipation of a digital circuit is a challenging problem. The transition density (switching activity) is a factor [12] that can be effective in calculating of the power dissipation. If the signal $x(T)$ makes $nx(T)$ transition in a specific time interval of length T and then the transition density of $x(T)$ is defined as:

$$D(x) = \lim_{T \to \infty} \frac{n_x{}^{(T)}}{T} \quad (2)$$

Transition probabilities $P_t(x)$ is the average fraction of clock cycles in a circuit when value of x at the end of the cycle is different from its initial value [11],[12],[14]. Probabilities are linked to power estimation. In a synchronous circuit (circuits with clock), the relationship between transition density and transition probability is:

$$D(x) \geq \frac{P_t(x)}{T_c} \quad (3)$$

Where T_c is circuit clock period, $D(x)$ and $P_t(x)$ are transition density and transition probability, respectively. If $x(t)$ is the logic signal at the gate output and $nx(T)$ is the number of transitions of x(t) between $-T/2$ and $+T/2$, then the average power defined as:

$$P_{av} = \lim_{T \to \infty} V_{DD} \cdot \frac{CV_{dd}\, n_x(T)/2}{T} \quad (4)$$

Where V_{DD} is the supply voltage and C is the logical gate capacitance load from the transistor-level description. The critical point about dynamic power calculation using above method is that results can't be affected by circuit internal delays. The dynamic power of each gate is defined as [12], [13]:

$$Dynamic\ Power = \frac{1}{2} \cdot f.Vdd^2.c.\alpha \quad (5)$$

Where f is the clock frequency, VDD is the supply voltage, C_i is capacitance load of the logic gate i and α_i is the switching activity of the logic gate i . As supply voltage and frequency of the circuit are the same for all of the circuit logic gates, we only compute switching activity and the capacitance load of the logic gates. To generate a matrix in order to illustrate the dynamic power heatmap presentation, we define G matrix as:

$$G = \begin{bmatrix} G_{11} & \cdots & G_{1n} \\ \vdots & \ddots & \vdots \\ G_{m1} & \cdots & G_{mn} \end{bmatrix} = \begin{bmatrix} c_{11}.\alpha_{11} & \cdots & c_{1n}.\alpha_{1n} \\ \vdots & \ddots & \vdots \\ c_{m1}.\alpha_{m1} & \cdots & c_{mn}.\alpha_{mn} \end{bmatrix} \quad (6)$$

G_{nm}=$c_{nm}.\alpha_{nm}$ is the matrix value from row m and column n and c_{mn} and α_{mn} are the capacitance and switching activity of the value of the matrix respectively. In order to provide G matrix, switching activity and capacitance load of each gate must be calculated.

B. Logic gates switching activity

We have provided switching activity and transition probability of each gate and node using the ACE 2 tool [16] from Verilog to routing tool (VTR). The ACE 2 tool provides open-source CAD tool for calculating logic gates switching activity. To provide the switching activities of each node of the circuit, Verilog HDL code of the testing circuits must be provided.

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TABLE I

ISCAS BENCHMARKS CHARACTERISTICS

Benchmark	Number of Inputs	Number of Outputs	Number of Total Gates
C17	5	2	6
C432	36	7	160
C5315	178	123	2307
S27	4	1	8
S298	3	6	75
S510	19	7	179
S1238	14	14	428
S5378	35	49	1004
S6669	83	55	2071
S35932	35	320	12204

TABLE II

C17 SIGNAL PROBABILITIES AND SWITCHING ACTIVITIES

Gates and nodes	Signal probability	Switching activity
G1	0.49560	0.19920
G2	0.48360	0.20440
G3	0.49500	0.19900
CLOCK	0.50000	2.00000
G4	0.55880	0.11893
G5	0.56400	0.22627
G6	0.75200	0.15437
Node [1]	0.38220	0.008024
Node [2]	0.37060	0.050244
Node [3]	0.38220	0.008024
Node [4]	0.24600	0.044761

The Odin II is used for converting Verilog Hardware description language into a BLIF netlist. Another step for providing activities is to use ABC. The ABC perform logic optimization and technology mapping. Activity information generated by the ACE tool consists of two values 1- the signal probability 2- switching activity. The switching activity generating process is shown in the Fig. 3. For example, the c17 circuit logic gates and nodes switching activities and signal probabilities are shown in the Table II. According to Table II, the switching activities of each gate are significantly higher than the nodes activity. Accordingly, we consider that the dynamic power of the logic gates is considerably higher than the dynamic power of the routing of the circuit. Although the dynamic power of the routing may have an effect in the final heatmaps, but in our case of study, we only consider dynamic power of the logic gates and the grids of the ASIC, which we explained in the last section, to provide estimated dynamic power heatmaps.

C. Logic gates capacitance load

In order to estimate the capacitance load, we must focus on the capacitance modeling of CMOS circuit and capacitance extraction. In [17], the authors present an equivalent load capacitance model computed from internal and interconnection capacitance. These capacitances can be calculated according to the designing factors, such as W/L (technology parameters). For logic-level description, there is only access to the logic gates input and output nodes. Consequently, to provide the capacitance factor of the G matrix, the output capacitance of the logic gates must be calculated. Fig. 1 shows the output capacitance of the c17 circuit logic gates (c1 to c6). We use the HSPICE tool in order to extract and calculate logic gates output capacitance to compute output capacitance more accurately. With the output capacitance calculated for each logic gate, c_{nm} factors from the G matrix can be determined.

D. Power heatmaps

As switching activity and capacitance load of each gate are provided, G matrix can be generated. Fig. 4 shows the generated heatmap for the C17 circuit using G matrix. We have used the seaborn library (written in Python) and the MATLAB in order to generate heatmaps. To generate secondary heatmaps with Trojan included, we insert Trojans to our initial ASIC grids and regenerate the heatmap again. Any change in the secondary - heatmaps emerges easily. In order to scan power heatmaps for Trojan detection, we have applied Image processing algorithms using the OpenCV library. OpenCV (open-source computer vision library) is an open-source computer vision software library. In the OpenCV, accessing to each pixel of the images and the images matrix, make it possible to examine provided heatmaps and propose an innovative approach to detect Hardware Trojans. The Default image matrix is defined as RGB color space. HSV (Hue, Saturation and Value) is another color model available that is widely used in place of the RGB color model [18]. Hue is the representation of the similarity of areas to perceived colors, which are red, yellow, green and blue. Saturation represents the colorfulness of an object to its own brightness. The HSV color model is more useful for tracking a specific object, which has specific colors combination, and thresholding the object in an image. Since hardware Trojans can be defined as an object in the heatmaps image, we use the HSV color model to track Trojans. The RGB heatmaps model generated in the last sections must be converted to the HSV color model. With this approach, overheated spots in the chips can be detected in the heatmaps. In the ISCAS circuits provided heatmaps, the highest switching activity belongs to the input clock (which is set as 2 in the ACE 2 simulations). This highest switching activity causes big red areas around the input clock location in the heatmaps. We mask the input clock activity in the heatmaps. However, clock grids and its effects on the circuit will not be removed.

F.Implementing Trojans

Hardware Trojans may be placed in the different locations of the circuit. Detecting Trojans can be a challenging problem due to the different sizes of Trojans and the power they consume. In this section, we design virtual Trojans and place them in different locations of the testing circuits. In [6], the authors have designed Trojans with local power density varying between a specific power consumption interval. As we have provided dynamic power heatmaps according to the G matrix, we design Trojans that change the switching activities of the Trojan included blocks. We evaluate our model for small Trojans that increase he switching activities of the blocks, between 0.1 to 1 percent. The switching activity has a direct relationship to the dynamic power. According to previous sections switching activity of the gates can be defined as:

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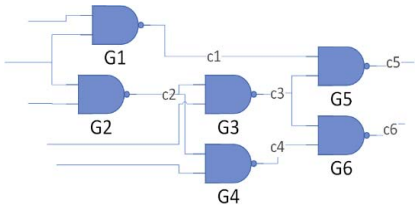

Fig 1. c17 circuit logic gates connection. Output capacitances are mentioned

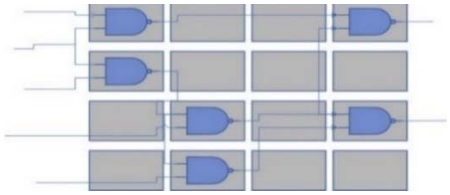

Fig 2. The c17 circuit placement in a ASIC with 16 grids

$$\alpha = \frac{P}{\frac{1}{2}.f.V_{dd}^{2}.C} \tag{7}$$

According to the above formula, if the switching activity increase, the dynamic power will increase correspondingly. Consequently, we design Trojans that affect switching activity and increase it between 0.1 to 1 percent. It means that these Trojans increase the dynamic power of the block by 0.1 to 1 percent, but as there are many blocks in the circuit, the final effect on the circuit power consumption is less than 0.1 percent. In the results section, we show that our detection method can detect small Trojan with very small power consumption. (a small change in the switching activity). In the last section, we also provided a dynamic power heatmap. With placing Trojans in the circuit and providing secondary heatmaps for Trojan included circuits, it is possible to detect Trojans with scanning heatmaps using image processing techniques.

E. Trojan detection

Extra activity in the Trojan included parts increases the switching activity in that location and has an effect on the functionality of the circuit. Consequently, the power and temperature of the circuit, especially near the Trojans' location, increase. This change of functionality may have effect on each node of the circuit switching activity. These changes in the circuits directly impact on the generated heatmaps from the circuit. In this paper, we have implemented gate-level hardware Trojans that have effects on the power and the functionality of the circuit. Fig. 5 compares the heatmaps generated from the c17 circuit with and without Trojan included. In order to generate a secondary heatmap for the c17 circuit in Fig. 5, we have inserted a gate-level Trojan that increases the switching activity of the grid by 20 percent. We place the gate-level Trojan in the G6 block (according to Fig. 1) of the c17 circuit. Differences between ideal and secondary heatmaps are apparent in the figure. The parts of the chip that contain Trojan appear more reddish than the other parts because of more switching activity. We have proposed image processing techniques in order to detect Trojans in the secondary heatmaps. Two main noticeable changes that emerge in the secondary heatmaps are 1- change overall heatmap appearance (Although some parts of the heatmaps may not change) and 2- appearing red spots around Trojans' locations. These spots have more switching activity in

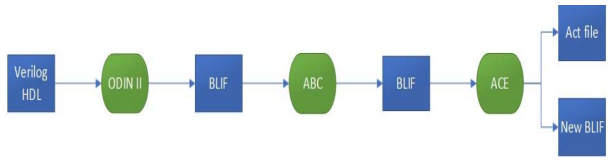

Fig 3. The switching activity calculation flowchart in ACE 2

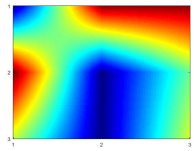

Fig 4. The c17 power heatmap

the G matrix, which is caused by Trojans activation triggers and Trojans activation. In order to track chip activity, we provide frame by frame heatmaps scanning and compare secondary heatmaps with an ideal heatmap in the long-term IC -run. According to our generated heatmaps using the switching activity technique, spots with more switching activity appear more reddish in the heatmaps. These red spots around Trojans' location in the heatmap emerged when activity of the Trojan in significantly higher rather than other parts of the chip. We use thresholding to segment more reddish spots using image HSV matrix to detect spots with higher switching activity. If Trojans' designers design the Trojans more creative, red spots around the Trojans location may not emerge. Fig. 5 illustrate the c17 Trojan included reddish spots masking using the thresholding technique. In the digital image processing, thresholding is a method for segmenting the image. If the Trojan designers consider ideal heatmaps of the circuit, they can design stealthy Trojans with lower switching activity. Consequently, secondary heatmaps will not have significant changes around the Trojans' location. To solve this problem, we apply the SSIM algorithm, and we will describe our approach to detect small Trojans that may not appear considerably in the heatmaps, but change overall circuits functionality. Trojans that cause changes in the circuit functionality are harder to detect as they may impact every location of the chip, not a specific area. These effects will be appearing in the dynamic power heatmaps. However, the difference between heatmaps may be negligible or significant, according to the Trojan design and activation model. These kinds of Trojans are harder to detect. Fig. 6 shows the c17 circuit heatmaps negligible differences after implementing small Trojan. To detecting these kinds of Trojans, we have used the SSIM method to calculate ideal and secondary heatmaps (which include Trojans) similarity and find the most suspicious locations. The SSIM can be used for measuring the similarities between the two images. In the SSIM approach, the system separates similarity to 3 different tasks 1-luminance 2-contrast, 3-structure. Comparison results in these three tasks make the final similarity measure [10],[19]. The Structural similarity index between signal x and y is defined as:

$$SSIM(x,y) = [l(x,y)].[c(x,y)].[s(x,y)] \tag{8}$$

Where $l(x,y)$, $c(x,y)$, $s(x,y)$ are the luminance, contrast and structure comparison between x and y, respectively. x and y are two different non-negative image signals. We use MATLAB to apply the SSIM method. It is expected 100 percent similarity-

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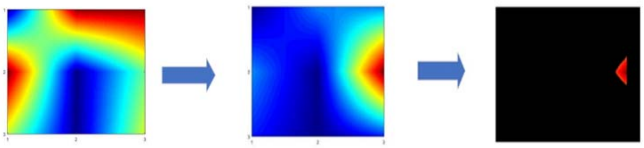

Fig 5. Trojan effect on the c17 heatmap. Trojan location is masked using thresholding technique

between power heatmaps if there are no Trojans included. With more functionality change in the circuit, the similarity percentage between heatmap may be descended. With testing this method, we can provide similarities percentages between ideal and Trojan included heatmaps for testing circuits. Examining these percentages can help to find if there are Trojans included or not. Also, changed locations in the secondary heatmaps can be emerged using the thresholding and the SSIM techniques. For example, Fig. 6 illustrates two heatmaps with a 95 percent similarity. Changed parts are mentioned and masked. According to the similarity percentage between heatmaps and thresholding technique, reliable Trojan detection with high accuracy can be performed. There are some errors in the algorithm that may be appeared by different noise sources in the heatmaps. Although, there are fewer sources of noise in the heatmaps than realistic power simulation of the circuit. As we have calculated the power of each part of the circuit and generate the heatmaps with the computer, there are less noises available in our method. The main error in Trojan detection using thermal maps and SEM imaging [7-9] is that thermal and image noises can have destructive effects on the final result. In order to compare our approach with SEM imaging and thermal maps scanning method, we add some various noises to the heatmaps to show that noises impact the final detection accuracy with the SSIM approach. Although there are several nonlinear filters such as the Median filter to denoising the images, but the main point is that these filters only reduce the noises and may cause a blurry effect on the image. In the next section, we describe our results, and we compare the results between the noisy heatmaps and the heatmaps without noise.

III. SIMULATION RESULTS

In this section, we evaluate our approach for different ISCAS benchmarks. We insert various types of Trojans with varying effects on the power and switching activity of the logic gates. We apply our proposed method to detect Trojans in the circuits. In our experiments, we compare the results with and without image noise to show that our suggested method can reach high accuracy rather than thermal scanning and SEM imaging. First, we insert various Trojans types with different effects on the power consumption and switching activity to observe the Trojans' effects on the heatmaps similarity percentage. We use SSIM to calculate the heatmaps similarity. The inserted Trojans increase the switching activities of the blocks, between 0.1 to 1 percent. The heatmaps similarity percentage with mentioned various Trojans are described in the Table III. In the next step, in order to evaluate our method, we insert Trojans in the circuits randomly in the different locations of the circuit's locations. As mentioned above, we design Trojans to increase the switching activity of the circuits blocks between 0.1 to 1 percent. With the SSIM and thresholding techniques, we try to locate the changed parts and mask them. These masked locations are the suspicious

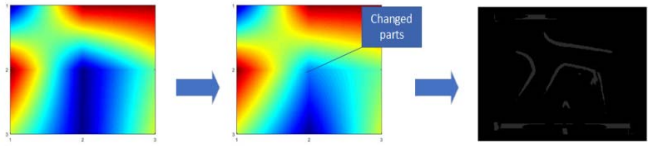

Fig 6. Small Trojans effect on the heatmap. The changed part is mentioned and masked in the figure using the thresholding technique

TABLE III

CIRCUIT'S HEATMAPS SIMILARITY PERCENTAGE FOR DIFFERENT SWITCHING ACTIVITY INCREASE

Benchmark	Similarity percentage for different Trojans					
	0%	*0.1%*	*0.2%*	*0.5%*	*0.8%*	*1%*
C17	100	99.97	99.17	97.63	92.58	83.87
C432	100	99.54	98.94	97.70	91.59	81.55
C5315	100	98.62	97.50	96.03	90.25	81.24
S27	100	99.96	98.66	96.92	92.54	82.56
S298	100	98.34	97.36	95.58	89.04	79.33
S510	100	97.62	95.68	93.11	88.34	77.81
S1238	100	95.23	92.51	88.58	84.27	74.25
S5378	100	95.26	91.82	85.20	83.59	73.65
S6669	100	94.74	88.57	85.35	81.26	69.65
S35932	100	94.15	89.41	82.59	79.65	61.96

locations of the circuits, which may include hardware Trojans. With inserting 100 different Trojans for each circuit and in the different locations of the circuit, we apply our method to detect Trojans. We record the average true decision (percentage of chips with Trojan inside and have been detected) and average missed (percentage of chips with Trojan inside but have not been detected). In the next step, in order to compare our proposed method with the SEM imaging and the thermal scanning, we add a different source of image noises (Gaussian, Salt-and-pepper, Quantization, Anisotropic and periodic noise) to the heatmaps to show that image noises can have effects on the SSIM technique results. Then we record the average true decision and average missed again, for the noisy heatmaps. According to Table III, similar heatmaps without noise are 100 percent similar. Still, we also recorded the average false alarm (circuit without Trojan included but reported as Trojan included circuits) for noisy heatmaps to show the noises' effects on the final result and comparing our approach with the SEM imaging. Table IV reports the average true decisions, average missed and average false alarm for each circuit in 100 different tests for various Trojans. The results for the heatmaps with noise are also reported in the Table IV. The average detection accuracy comparison between noisy heatmaps and noiseless heatmaps for each circuit is shown in the Fig. 7. The average detection accuracy comparison between noisy heatmaps and noiseless heatmaps (proposed method) for each circuit is shown in the Fig. 7. According to the results, we summarize a comparison between our proposed method with the other Trojan detection methods in different criteria in the Table V.

IV. CONCLUSION AND FUTURE WORK

In this paper, we have presented a novel approach to Hardware Trojan detection and classifying Trojan included circuits with 97 percent average accuracy for small Trojans, which consume power less than 0.001 of the total power consumption. First, we generated power heatmaps with the -

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TABLE IV

AVERAGE DETECTION ACCURACY PERCENTAGE IN 100 TESTS

Benchmark	true decision (without noise)	missed (without noise)	true decision (with noise)	missed (with noise)	false alarm (with noise)
C17	100	0	100	0	0
C432	99	1	95	5	3
C5315	98	2	94	6	6
S27	100	0	99	1	2
S298	98	2	90	1	9
S510	96	4	88	12	8
S1238	95	5	84	16	15
S5378	96	4	80	20	22
S6669	94	6	74	26	27
S35932	93	7	69	31	36

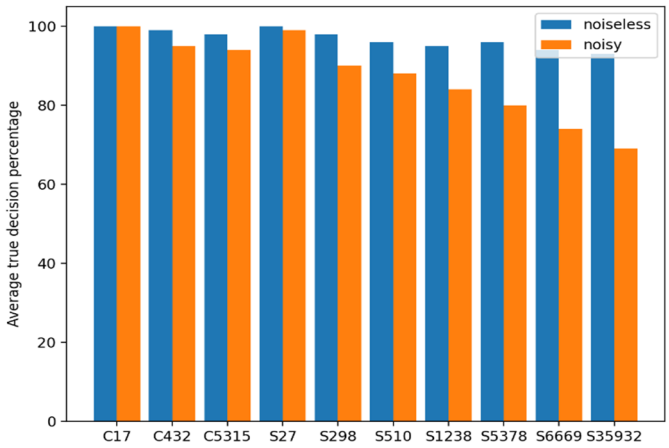

Fig 7. Average detection accuracy with the proposed method (blue bars) and with noises (orange bars) for various types of Trojans with a different effect on heatmaps

TABLE V

COMPARING OTHER METHODS WITH OUR PROPOSED TECHNIQUE

Method	Sample required	Time needed	Noise effect	Trojans detected
Reverse engineering	50<	Months	-	All
Thermal scanning	<10	Hours	High	96%
SEM imaging	1	Hours	High	95%
proposed technique	1	Hours	Low	97%

switching activity method, and we applied image processing techniques on the generated heatmaps. Obtained simulation results showed the efficiency and the high accuracy of the proposed method. Future work will mainly cover developing the SSIM and other image processing techniques and applying machine learning classification algorithms such as the neural networks, which can improve our approach and accuracy.

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